

FALL 2026



EE3003 Analog and Digital Communications

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FALL 2026

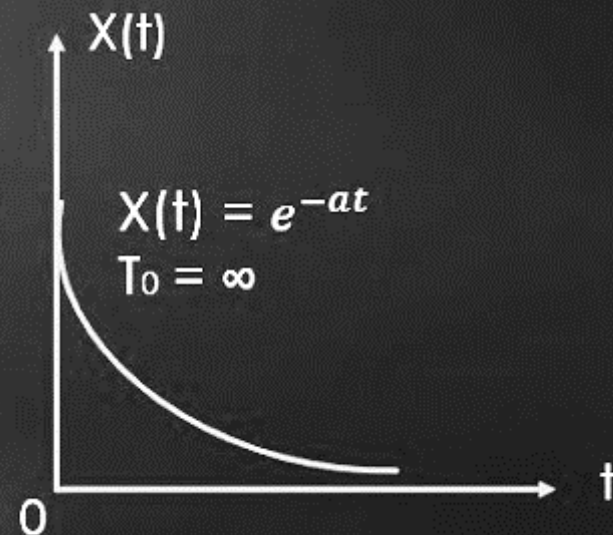
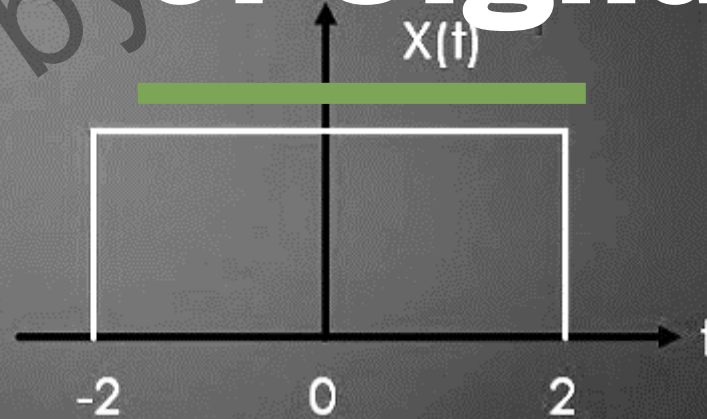
Objectives

- Classification of Signals
- Signal Energy and Energy Spectral Density (ESD)
- Signal Power and Power Spectral Density (PSD)

Signal & Signal Classification

- I. CONTINUOUS-TIME & DISCRETE-TIME SIGNALS.
- II. DETERMINISTIC & RANDOM SIGNALS.
- III. EVEN & ODD SIGNALS.
- IV. PERIODIC & NON-PERIODIC SIGNALS.
- V. ENERGY & POWER SIGNALS.
- VI. ANALOG & DIGITAL SIGNALS.
- VII. REAL & COMPLEX SIGNALS.

Classification of Signals



Signal Energy

- A signal with *finite energy* is an *energy signal*.

➤ Area under $|g(t)|^2$ [for complex valued signals],

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

➤ A signal is an energy signal if,

$$\int_{-\infty}^{\infty} |g(t)|^2 dt < \infty$$

❖ A necessary condition for the energy to be finite is that the amplitude $\rightarrow 0$ as $|t| \rightarrow \infty$.

Signal Energy using Parseval's Theorem

- To determine the signal energy from either the **time-domain specification $g(t)$** or the **frequency-domain specification $G(\omega)$** of the same signal.

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

$$E_g = \int_{-\infty}^{\infty} g(t) g^*(t) dt = \int_{-\infty}^{\infty} g(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) e^{-j\omega t} d\omega \right] dt$$

Interchanging the order of integration yields,

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) \left[\int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt \right] d\omega$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) G(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

$$G(\omega) = \mathcal{F}[g(t)]$$

$$g(t) = \mathcal{F}^{-1}[G(\omega)]$$

$$g(t) \iff G(\omega)$$

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} d\omega$$

Example (Parseval's Theorem)

Verify Parseval's Theorem for the signal, $g(t) = e^{-at}u(t)$, for $a > 0$.

We have in **time-domain**,

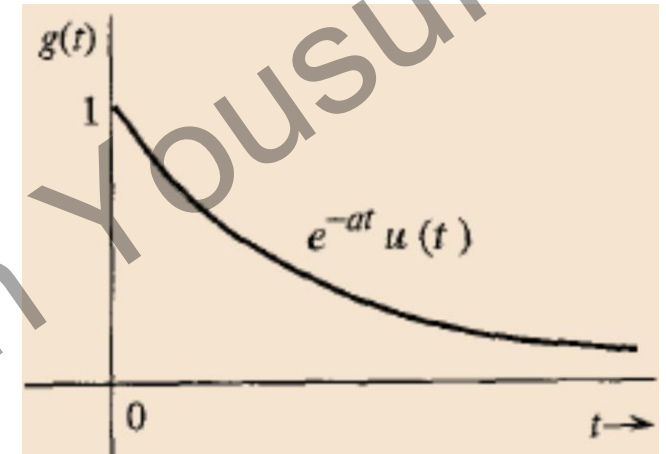
$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

$$E_g = \int_0^{\infty} e^{-2at} dt = \left[\frac{e^{-2at}}{-2a} \right]_0^{\infty} = \frac{1}{2a}$$

We now determine E_g from the signal **spectrum** $G(\omega)$ given by,

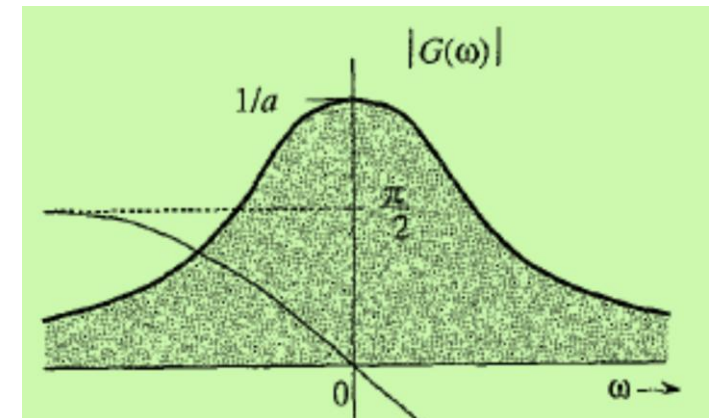
$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega$$

$$E_g = \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-\infty}^{\infty} = \frac{1}{2\pi a} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \frac{1}{2a}$$



$$G(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t} dt$$

$$G(\omega) = \frac{1}{a + j\omega} \quad a > 0$$



Energy Spectral Density (ESD), $\Psi_g(\omega)$

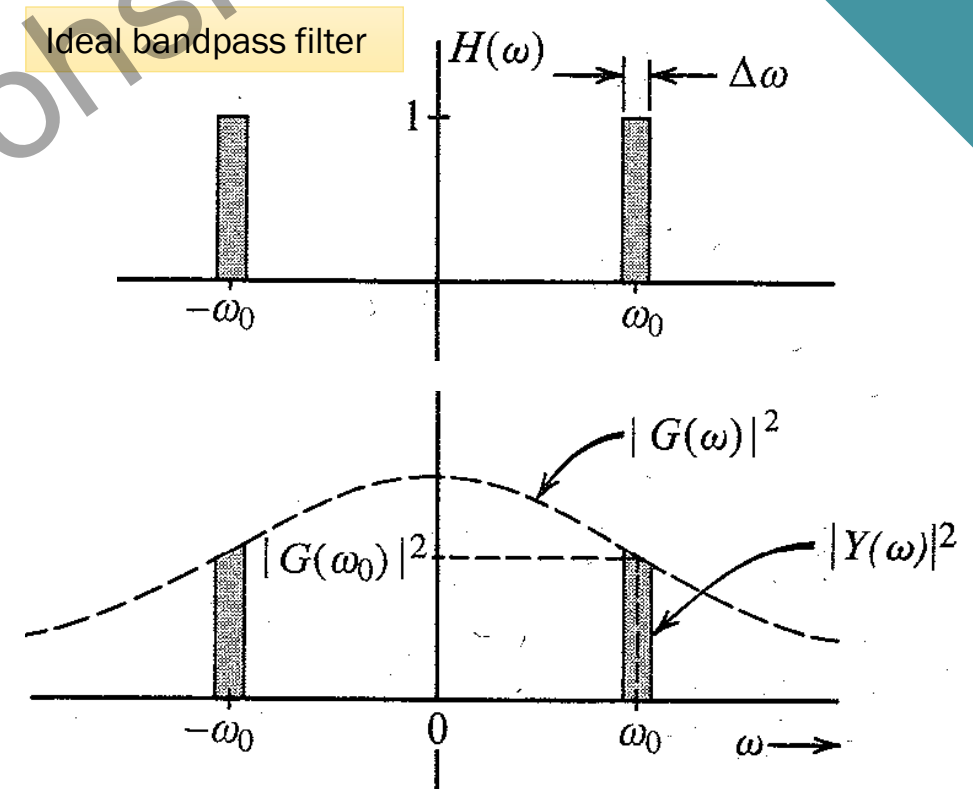
$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

The energy of a signal $g(t)$ is the result of energies contributed by all the **spectral components** of the signal $g(t)$.

- The contribution of a spectral component of frequency ω is proportional to $|G(\omega)|^2$.
- Consider a signal $g(t)$ applied at the input of an ideal bandpass filter, whose transfer function $H(\omega)$ is shown.
- If the filter output is $y(t)$, then its Fourier transform $Y(\omega) = G(\omega)H(\omega)$, and E_y , the energy of the output $y(t)$ is,

$$E_y = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)H(\omega)|^2 d\omega$$

Because $H(\omega)=1$ over the passband $\Delta\omega$, and zero everywhere else, the integral on the right-hand side is the sum of the two shaded areas in Fig. 3.36b, and we have (for $\Delta\omega \rightarrow 0$)



$$E_y = 2 \frac{1}{2\pi} |G(\omega_0)|^2 d\omega = 2|G(\omega_0)|^2 df$$

Energy Spectral Density (ESD), $\Psi_g(\omega)$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

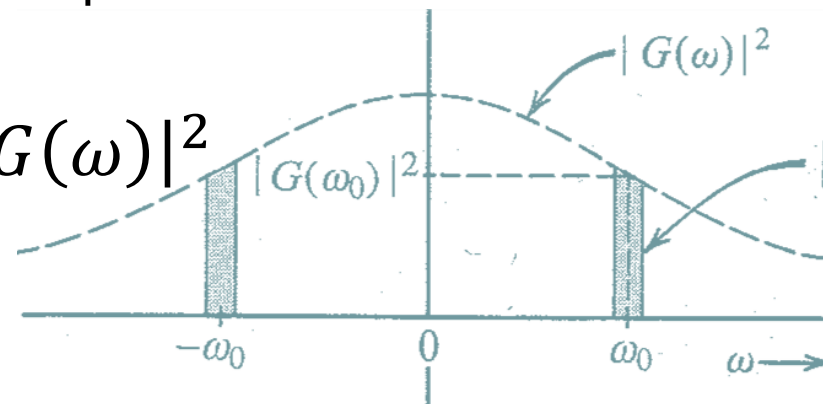
$$E_y = 2 \frac{1}{2\pi} |G(\omega_0)|^2 d\omega = 2|G(\omega_0)|^2 df$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi_g(\omega) d\omega$$

Thus $2|G(\omega)|^2 df$ is the energy contributed by the spectral components within the two narrow bands, each of width Δf Hz, centered at $\pm\omega_0$.

So, $|G(\omega)|^2$ is the **energy per unit bandwidth** (in hertz), also known as energy spectral density (ESD) of the spectral components of $g(t)$ centered at frequency ω .

$$ESD = \Psi_g(\omega) = |G(\omega)|^2$$



In the Example,

$$g(t) = e^{-at}u(t)$$

$$G(\omega) = \frac{1}{a + j\omega}$$

$$\Psi_g(\omega) = |G(\omega)|^2 = \frac{1}{\omega^2 + a^2}$$

Essential Bandwidth of a Signal

Bandwidth Estimation Procedure

- A time-limited signal has spectrum that extends to infinity. Practical signals have finite energy, so, the signal spectrum, $G(\omega)$ must approach 0 as $\omega \rightarrow \infty$.
- Most of the signal energy is contained within a certain band of B Hz, and the energy content beyond it is negligible and can be suppressed with little effect on the signal shape and energy.
- The bandwidth B is called **essential bandwidth** of the signal.
 - The criterion for selecting B depends on the **error tolerance** in a particular application and usually lies between **90 to 99 % of the signal energy**.

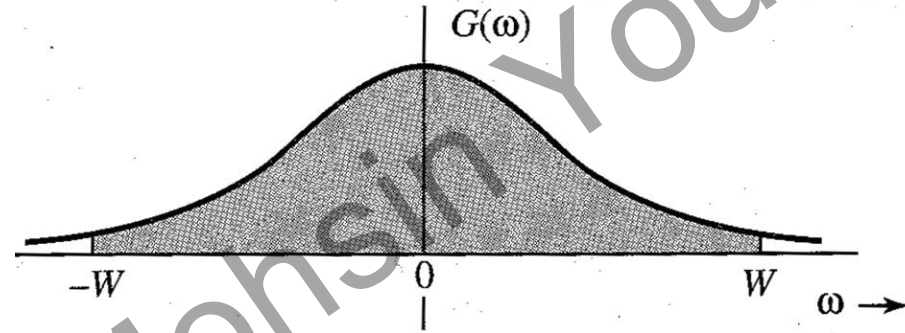
EXAMPLE 3.20 Estimate the essential bandwidth W rad/s of the signal $e^{-at}u(t)$ if the essential band is required to contain 95% of the signal energy.

In this case,

$$G(\omega) = \frac{1}{j\omega + a}$$

and the ESD is

$$|G(\omega)|^2 = \frac{1}{\omega^2 + a^2}$$



$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega = \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-\infty}^{\infty} = \frac{1}{2a}$$

$$\begin{aligned} \frac{0.95}{2a} &= \frac{1}{2\pi} \int_{-W}^W \frac{d\omega}{\omega^2 + a^2} \\ &= \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-W}^W = \frac{1}{\pi a} \tan^{-1} \frac{W}{a} \end{aligned}$$

$$\frac{0.95\pi}{2} = \tan^{-1} \frac{W}{a} \implies W = 12.706a \text{ rad/s}$$

This means that the spectral components of $g(t)$ in the band from 0 (dc) to 12.706 rad/s (2.02 Hz) contribute 95% of the total signal energy; all the remaining spectral components (in the band from 12.706 rad/s to ∞) contribute only 5% of the signal energy.*

Energy of the Modulated Signals

Let $g(t)$ be a baseband signal band-limited to B Hz. The amplitude-modulated signal $\varphi(t)$ is

$$\varphi(t) = g(t) \cos \omega_0 t$$

and the spectrum (Fourier transform) of $\varphi(t)$ is

$$\Phi(\omega) = \frac{1}{2}[G(\omega + \omega_0) + G(\omega - \omega_0)]$$

The ESD of the modulated signal $\varphi(t)$ is $|\Phi(\omega)|^2$, that is,

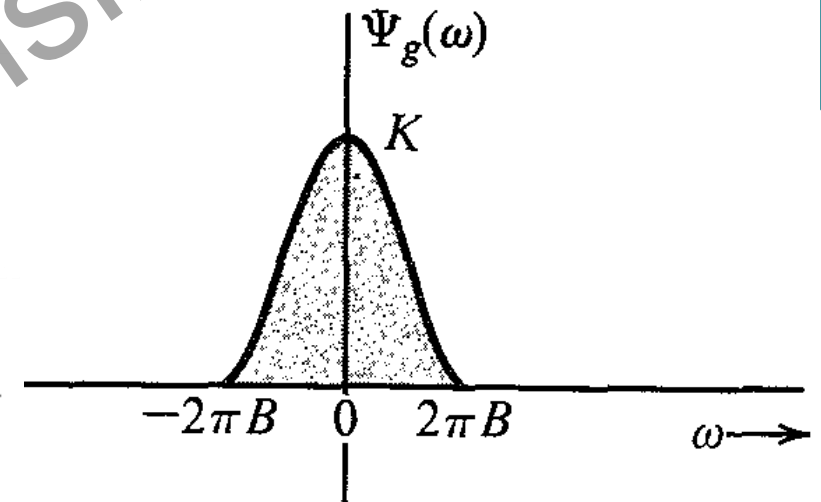
$$\Psi_\varphi(\omega) = \frac{1}{4}|G(\omega + \omega_0) + G(\omega - \omega_0)|^2$$

If $\omega_0 \geq 2\pi B$, then $G(\omega + \omega_0)$ and $G(\omega - \omega_0)$ are nonoverlapping (see Fig. 3.39), and

$$\Psi_\varphi(\omega) = \frac{1}{4}[|G(\omega + \omega_0)|^2 + |G(\omega - \omega_0)|^2] \quad (3.69a)$$

$$= \frac{1}{4}[\Psi_g(\omega + \omega_0) + \Psi_g(\omega - \omega_0)] \quad (3.69b)$$

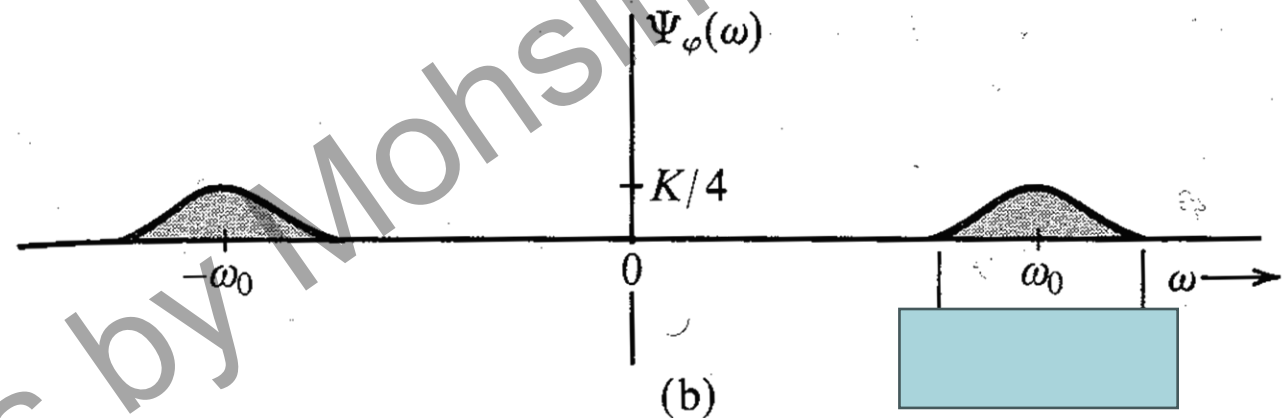
ESD of modulating signal $g(t)$



Energy of the Modulated Signals

ESD of modulated signal $\varphi(t)$

$$\begin{aligned}\Psi_{\varphi}(\omega) &= \frac{1}{4} \left[|G(\omega + \omega_0)|^2 + |G(\omega - \omega_0)|^2 \right] \\ &= \frac{1}{4} \left[\Psi_g(\omega + \omega_0) + \Psi_g(\omega - \omega_0) \right]\end{aligned}$$



The ESDs of both $g(t)$ and the modulated signal $\varphi(t)$ are shown in Fig. 3.39. It is clear that modulation shifts the ESD of $g(t)$ by $\pm\omega_0$. Observe that the area under $\Psi_{\varphi}(\omega)$ is half the area under $\Psi_g(\omega)$. Because the energy of a signal is proportional to the area under its ESD, it follows that the energy of $\varphi(t)$ is half the energy of $g(t)$, that is,

$$E_{\varphi} = \frac{1}{2} E_g \quad \omega_0 \geq 2\pi B \quad (3.70)$$

Energy of modulated signal is half the energy of baseband signal

Time Autocorrelation Function $\psi_g(\tau)$

& ESD $\Psi_g(\omega)$

A good measure of comparing two signals $g(t)$ and $z(t)$ is the correlation function $\psi_{gz}(\tau)$. The correlation of a signal $g(t)$ with itself is **autocorrelation** $\psi_g(\tau)$ given as,

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t + \tau) dt \quad (3.71a)$$

Setting $x = t + \tau$ in Eq. (3.71a) yields

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(x)g(x - \tau) dx$$

In this equation, x is a dummy variable and could be replaced by t . Thus,

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t \pm \tau) dt \quad (3.71b)$$

This shows that for a real $g(t)$, the autocorrelation function is an even function of τ , that is,

$$\psi_g(\tau) = \psi_g(-\tau) \quad (3.72)$$

Time Autocorrelation Function $\psi_g(\tau)$ & ESD $\Psi_g(\omega)$

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t + \tau) dt$$

We now show that the ESD $\Psi_g(\omega) = |G(\omega)|^2$ is the Fourier transform of the autocorrelation function $\psi_g(\tau)$. Although the result is proved here for real signals, it is valid for complex signals also. Note that the autocorrelation function is a function of τ , not t . Hence, its Fourier transform is $\int \psi(\tau)e^{-j\omega\tau} d\tau$. Thus,

$$\begin{aligned}\mathcal{F}[\psi_g(\tau)] &= \int_{-\infty}^{\infty} e^{-j\omega\tau} \left[\int_{-\infty}^{\infty} g(t)g(t + \tau) dt \right] d\tau \\ &= \int_{-\infty}^{\infty} g(t) \left[\int_{-\infty}^{\infty} g(\tau + t)e^{-j\omega\tau} d\tau \right] dt\end{aligned}$$

The inner integral is the Fourier transform of $g(\tau + t)$, which is $g(\tau)$ left-shifted by t . Hence, it is given by [time-shifting property in Eq. (3.30)] $G(\omega)e^{j\omega t}$. Therefore,

$$\mathcal{F}[\psi_g(\tau)] = G(\omega) \int_{-\infty}^{\infty} g(t)e^{j\omega t} dt = G(\omega)G(-\omega) = |G(\omega)|^2$$

This shows that

$$\psi_g(\tau) \Longleftrightarrow \Psi_g(\omega) = |G(\omega)|^2 \quad (3.73)$$

EXAMPLE 3.22 Find the time autocorrelation function of the signal $g(t) = e^{-at}u(t)$, and from it determine the ESD of $g(t)$.

In this case,

$$g(t) = e^{-at}u(t) \quad \text{and} \quad g(t - \tau) = e^{-a(t-\tau)}u(t - \tau)$$

Recall that $g(t - \tau)$ is $g(t)$ right-shifted by τ , as shown in Fig. 3.40a (for positive τ). The autocorrelation function $\psi_g(\tau)$ is given by the area under the product $g(t)g(t - \tau)$ [see Eq. (3.71b)]. Therefore,

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t - \tau) dt = e^{a\tau} \int_{\tau}^{\infty} e^{-2at} dt = \frac{1}{2a} e^{-a\tau}$$

we know that for a real $g(t)$, $\psi_g(\tau)$ is an even function of τ . Therefore,

$$\psi_g(\tau) = \frac{1}{2a} e^{-a|\tau|}$$

ESD of $\psi_g(\tau)$

$$\Psi_g(\omega) = \frac{1}{\omega^2 + a^2}$$

